

Cloud Computing Career Roadmap

Zero to Internship/Job-Ready — 100% Free Video Resources & Free Certificates

This is a complete, self-paced roadmap built for an absolute beginner who wants to go deep into cloud computing and land an internship or entry-level job. Every resource listed is **free** and, wherever possible, **video-based**. No paid courses are required. Total realistic timeline: **6-9 months** if you study 1.5-2 hours a day, 5-6 days a week — faster if you can dedicate more time.

How to use this roadmap: Follow the phases in order. Don't skip the hands-on labs and projects — recruiters care far more about what you've *built* than which videos you watched. Each phase lists the goal, free video course(s), free certificate/badge options, and what to build before moving on.

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PHASE 0 — Set Up & Fundamentals

Duration: 2-3 days | Goal: Get your tools ready

[Free video](#)

PHASE 1 — IT & Networking Foundations

Duration: 2 weeks | Goal: Understand how the internet, servers and networks actually work

You cannot understand the cloud without understanding networking. This is the #1 skipped step that causes people to memorize cloud services without understanding them.

[Free video courses](#)

[What you must understand by the end](#)

[Free certificate option](#)

CompTIA Network+ itself is paid, but you don't need the cert now — Professor Messer's free course covers 100% of the same syllabus for knowledge purposes.

PHASE 2 — Linux & Command Line

Duration: 1.5-2 weeks | Goal: Be comfortable in a Linux terminal — every cloud server runs on it

[Free video courses](#)

[Practice \(free\)](#)

PHASE 3 — Git, GitHub & Version Control

Duration: 4-5 days | Goal: Track code, collaborate, and build your public portfolio

[Free video courses](#)

[Practice](#)

PHASE 4 — Cloud Computing Core Concepts

Duration: 1.5-2 weeks | Goal: Understand cloud theory before touching any single provider

[Free video courses](#)

[Core topics to master](#)

PHASE 5 — Deep Dive: AWS (Primary Cloud Platform)

Duration: 4-5 weeks | Goal: Become genuinely job-capable on AWS, the most in-demand cloud

AWS has the largest market share and the most entry-level jobs/internships. Go deep here first, then add Azure/GCP awareness later (Phase 10).

Step 1 — AWS Cloud Practitioner (foundation, do this first)

Step 2 — Hands-on with real AWS Free Tier

Step 3 — Go one level deeper: Solutions Architect Associate content (optional but recommended)

Free certificate/badge options for AWS

PHASE 6 — Scripting for Cloud: Python + Bash

Duration: 2-3 weeks | Goal: Automate cloud tasks instead of clicking buttons

Free video courses

Practice

PHASE 7 — Infrastructure as Code: Terraform

Duration: 2 weeks | Goal: Provision cloud infrastructure with code, not clicks — a top hiring skill

Free video courses

Practice

Free certificate

[HashiCorp Terraform Associate study materials are free on HashiCorp's site](#) (the exam itself is paid — treat as a stretch goal later).

PHASE 8 — Containers: Docker & Kubernetes

Duration: 3 weeks | Goal: Package and orchestrate applications — essential in almost every cloud job posting

Free video courses

Practice

PHASE 9 — CI/CD & DevOps Basics

Duration: 2 weeks | Goal: Automate build-test-deploy pipelines

[Free video courses](#)

[Practice](#)

PHASE 10 — Multi-Cloud Awareness: Azure & Google Cloud

Duration: 2 weeks | Goal: Speak the language of the other two major clouds — many job posts ask for "AWS/Azure/GCP"

[Microsoft Azure — free resources](#)

[Google Cloud — free resources](#)

Goal for this phase: understand the AWS-equivalent service names on Azure/GCP (EC2≈Azure VM≈Compute Engine, S3≈Blob Storage≈Cloud Storage, IAM≈Azure AD≈Cloud IAM, etc.) — you don't need to be an expert in all three.

PHASE 11 — Cloud Security Basics

Duration: 1 week | Goal: Security is asked about in almost every cloud interview

[Free video courses](#)

[Core topics](#)

PHASE 12 — Hands-on Projects Portfolio

Build these throughout Weeks 10-24 | Goal: This is what actually gets you hired, not certificates

Build at least 5 of these, document each with a README + architecture diagram, and push to GitHub

Tip: Write a short blog post (on dev.to or [Hashnode](https://hashnode.com), both free) for each project explaining what you built and why — this becomes powerful proof of communication skills.

PHASE 13 — Resume, GitHub Profile & LinkedIn

Duration: 3-4 days | Goal: Package everything so recruiters can find and trust you

PHASE 14 — Applying & Interview Prep

Ongoing from Week 23 | Goal: Convert skills into an internship/job offer

Where to apply

Free interview prep

Appendix A — Free Certificates You Can Actually Earn

Provider	What you get	Cost
AWS Skill Builder	Digital course completion certificates + skill badges	Free
AWS Educate	Cloud badges + credits for students/learners	Free
Microsoft Learn	Learning path "trophies" + free AZ-900 voucher via Cloud Skills Challenges (occasional)	Free
Google Cloud Skills Boost / Arc42	Skill badges from labs + arcade points/badges	Free
IBM SkillsBuild	Cloud computing courses with completion certificates + digital badges	Free
freeCodeCamp	Its own certifications (general dev/scripting, not cloud-specific) + free cloud videos	Free
HashiCorp Developer / Terraform	Study material and practice (exam itself paid)	Free study
Great Learning Academy	Short courses with a certificate of completion	Free
Simplilearn SkillUp	Free courses with a certificate of completion	Free

Honest note: The flagship paid exams (AWS CCP/SAA, AZ-900, Google Associate Cloud Engineer) each cost roughly \$70-165 and are NOT free. Everything above gets you real, verifiable free credentials while you learn — treat one paid certification as a later investment once you've validated your skills with projects and are closer to applying, ideally after saving a little or via a student/voucher discount.

Appendix B — YouTube Channels Worth Subscribing To

Appendix C — Communities & Free Practice Platforms

Final advice: Certificates open doors slightly, but projects + the ability to explain your architecture choices in an interview is what actually gets offers. Build in public, push code weekly, and revisit this roadmap phase by phase. Good luck — you've got this.